Maxx Steele Mechanical Manual

Disassembly & Reassembly

Maxx Steele Mechanical Manual

Maxx-Steele-1984-Robot contributors

2026-11-06

Contents

Maxx Steele Mechanical Manual **2**
- What this guide covers . . . 2
- Before you start . . . 3
- How to use this manual . . . 3

Chapter 1 — Disassembly **4**
- Preparation . . . 4
- Remove stabilizers . . . 4
- Open the main case . . . 4
- Remove the head / remote holder . . . 4
- Separate the back shell . . . 4

Chapter 2 — Reassembly **5**
- Close the main case . . . 5
- Reinstall the head / remote holder . . . 5
- Reinstall stabilizers . . . 5
- Finish . . . 5

Chapter 3 — Chassis photo reference **6**
- Workshop teardown sequence . . . 6
- Cyberia Makerspace — workshop session . . . 22
- Internal autopsy . . . 27
- Exterior and collection . . . 34
- Case and packaging . . . 49
- Body artwork . . . 53

Maxx Steele Mechanical Manual

This manual describes how to **disassemble and reassemble** the 1984 CBS Toys / Ideal **Maxx Steele** robot chassis for repair, inspection, and maintenance.

It is a community edition of a workshop guide originally written by **Mike McCarty** (preserved in `Sources/Maxx-Steele-Disassembly-Guide.docx`).

PDF (latest): `Maxx-Steele-Mechanical-Manual.pdf` — rebuilt when `.md` or cover images change (`tools/build_mechanical_manual_pdf.py` locally; GitHub Actions on push to `main`).

What this guide covers

Document	Audience	Focus
This guide	Owners, repairers	Case opening, head module, stabilizers, reassembly
`Docs/Technical/`	Programmers, reverse engineers	Bytecode, ROM, I/O, cartridge authoring
`Docs/User/`	Owners	Setup, operation, games, maintenance (markdown + PDF)
`Chassis/Manual/`	Owners	Archival factory PDF scans
`Chassis/Photos/`	Reference	All chassis photos (embedded in Ch 3 by folder)
`Chassis/Artwork/`	Reference	Body logos and graphics (listed in Ch 3)

Before you start

- **Power off** and **remove the battery** before opening the chassis.
- Work on a soft surface; lay Maxx **face down** when removing the back shell.
- **Do not strain** the wires between the back shell and the body (remote holder, display, RF).
- Keep screws with their hardware (stabilizers, case, head plate).

Module-level PCB work is documented under `Mainboard/`, `Face/`, `Receiver/`, `Power/`, and `Transmitter/`.

How to use this manual

1. Chapter 1 — Disassembly
2. Chapter 2 — Reassembly
3. Chapter 3 — Chassis photo reference — all `Chassis/Photos/` and `Chassis/Artwork/` images by folder

Regenerate the photo chapter after adding chassis images:

```
python3 tools/gen_mechanical_chassis_photos.py
```

Hardware media index (photos, sounds, firmware, schematics): `Hardware-Assets.csv`. Regenerate with `python3 tools/gen_hardware_assets_csv.py`. Manual classifications are under human review (#42).

For programming and electrical reference after reassembly, see `Docs/Technical/README.md`.

Chapter 1 — Disassembly

Follow these steps to open the Maxx Steele chassis for inspection or repair. Complete them in order.

Preparation

1. Move Maxx’s arms to the **fully up** position.
2. Turn Maxx **off** and **remove the battery**.

Remove stabilizers

3. Remove Maxx’s **front stabilizer**:
   - Unscrew the two screws holding the stabilizer in place.
   - Gently pry the two fingers that slide under the body with a **wide-blade screwdriver**.
   - Place the stabilizer and screws together so they are not separated.
4. Repeat step 3 on the **rear stabilizer**.

Open the main case

5. Lay Maxx **face down** and remove the **twelve screws** holding the back of the case to the front.

The community source uses twelve in the removal steps and fourteen in the reassembly steps. Both numbers are copied as written. Chapter 8 records that pair. No single count was substituted.

Remove the head / remote holder

6. Remove the **remote control holder** from Maxx’s head:
   - Remove the screw from the front plate (the plate holding his “eyes”).
   - Remove the front plate and the sponge pad behind the eyes.
   - Press gently on the **tang at the rear** of the remote control holder and lift the holder away from the head.
   - **Take care not to strain** the wires going to the remote control holder. Lay the holder to the side.

Separate the back shell

7. Remove the **two case screws** that were hidden by the remote control holder.
8. Remove the **back of the case**, again being careful of the wires from the inside body to the back.
9. Press gently on Maxx’s **“armpits”** so the case can clear his **“shoulders”**.

You now have access to the interior. For board-level work, see `Mainboard/` and `Face/`.

Next: Chapter 2 — Reassembly

Chapter 2 — Reassembly

Reverse the disassembly sequence. **Do not rely on screws to pull the case halves together** — align and press the shell closed first.

Close the main case

1. Place Maxx on his **front** and carefully position the **back shell**, aligning all screw holes.
2. Press gently on Maxx's **"armpits"** so they clear the **"shoulders"**. Press the case halves together **before** installing screws.
3. Reinstall the **fourteen screws** holding the case back to the front using the **thread-engagement technique**. The disassembly text says twelve screws come out. Both numbers are the community source's numbers. See Chapter 8.
   - Insert one screw into its hole.
   - Turn it **counterclockwise** until the threads engage (you will feel the screw move out, then suddenly drop in with a slight thump).
   - Repeat the counterclockwise start until you are confident the threads are engaged.
   - Tighten only until the screw **just resists** further turning — this helps avoid stripped threads.

Reinstall the head / remote holder

4. Reinstall the three **"eyes"** and the **foam pad** in the front plate on the top of the head.
5. Place the **remote control holder** in position, then install the front plate and its securing screw.

Reinstall stabilizers

6. Replace the **rear stabilizer**: slide it onto the body, press down and forward until the under-body fingers engage, then install the two screws.
7. Turn Maxx onto his back and repeat step 6 for the **front stabilizer**.

Finish

8. Stand Maxx upright and install the **battery**.

Previous: Chapter 1 — Disassembly · **Next:** Chapter 3 — Chassis photo reference

Chapter 3 — Chassis photo reference

All photos and artwork under `Chassis/`, grouped by source folder.

Workshop teardown sequence

Primary disassembly walkthrough (`IMG_2116–IMG_2131`). Archive: `Chassis/Photos/Disassembly/`.

Figure 1: IMG_2116.JPG

Figure 2: IMG_2117.JPG

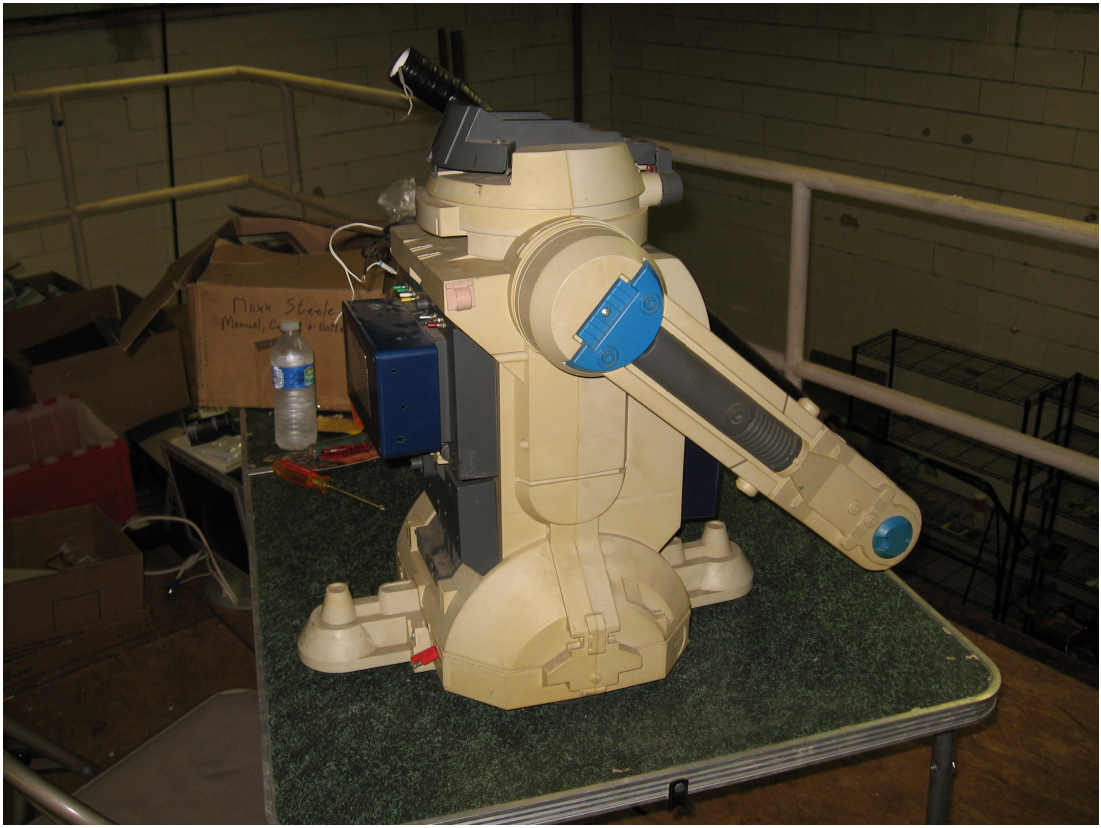

Figure 3: IMG_2118.JPG

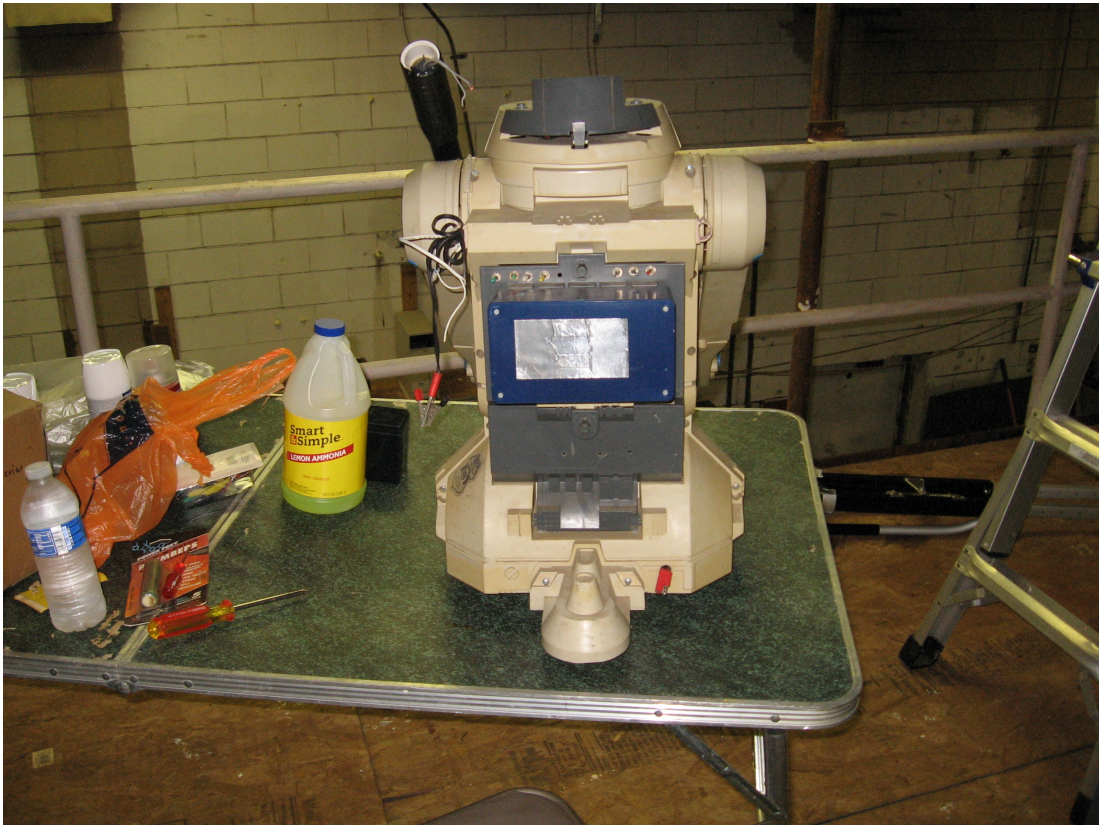

Figure 4: IMG_2119.JPG

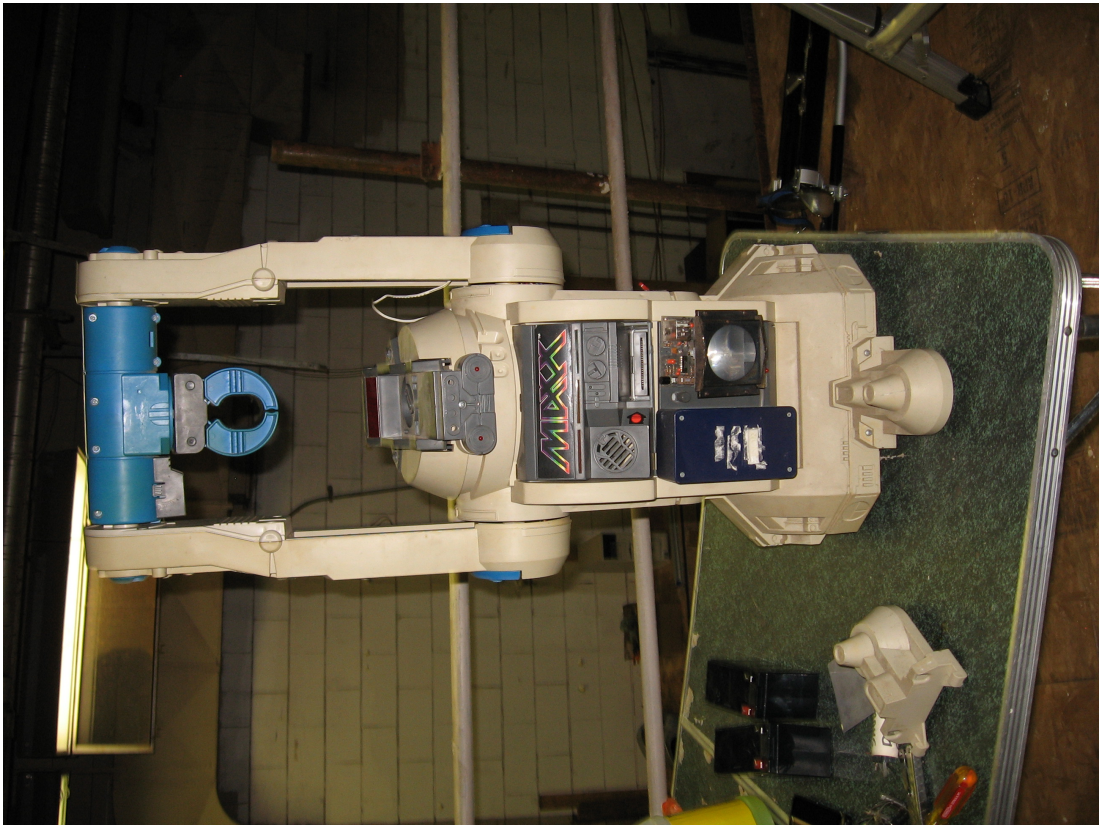

Figure 5: IMG_2120.JPG

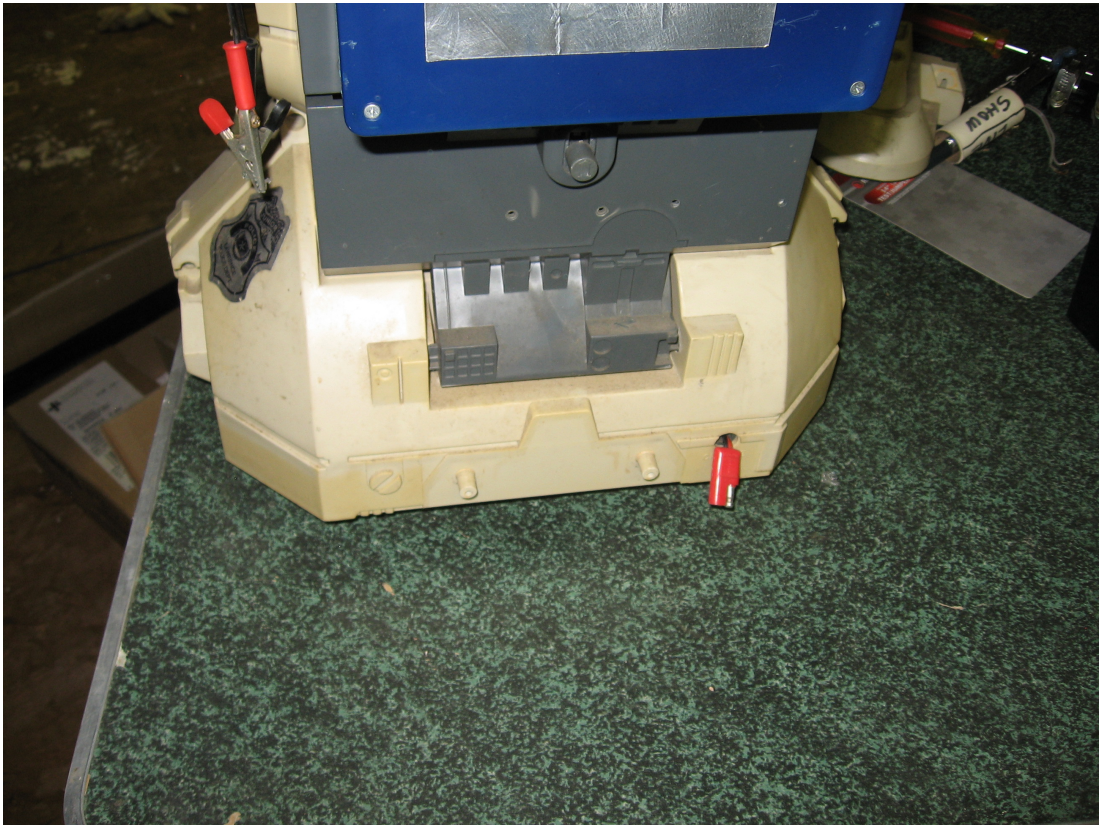

Figure 6: IMG_2121.JPG

Figure 7: IMG_2122.JPG

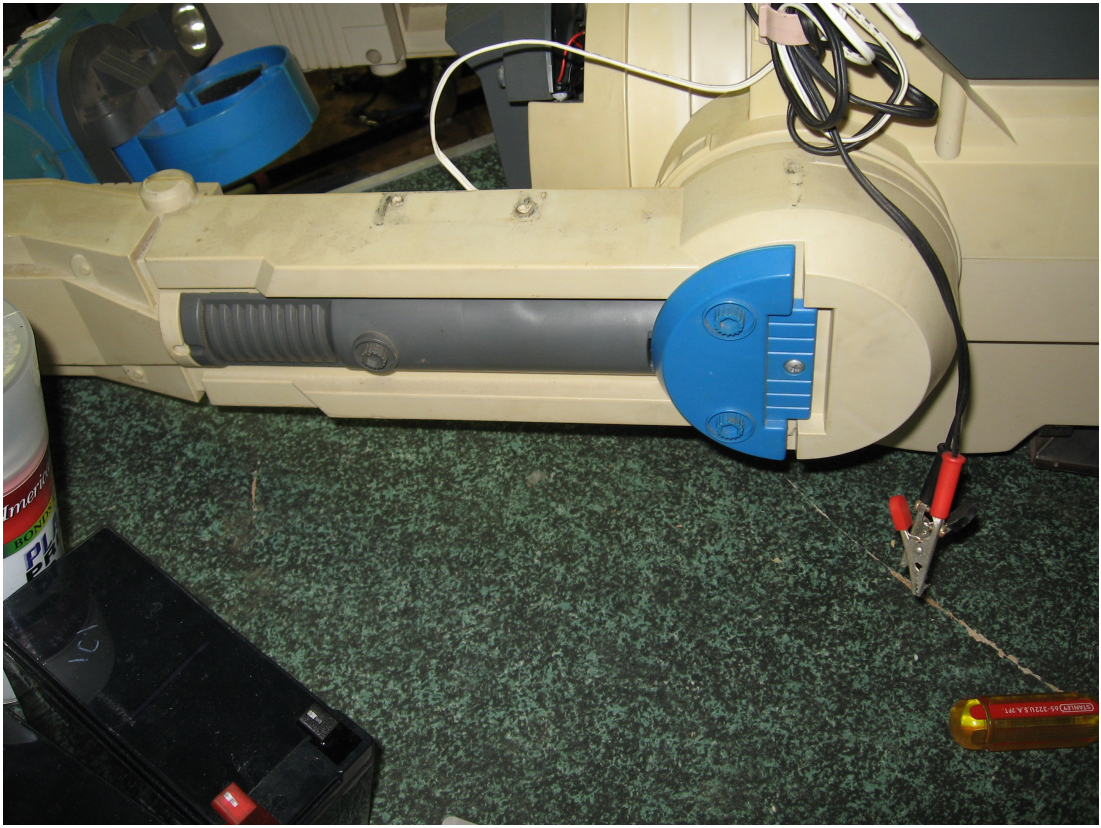

Figure 8: IMG_2123.JPG

Figure 9: IMG_2124.JPG

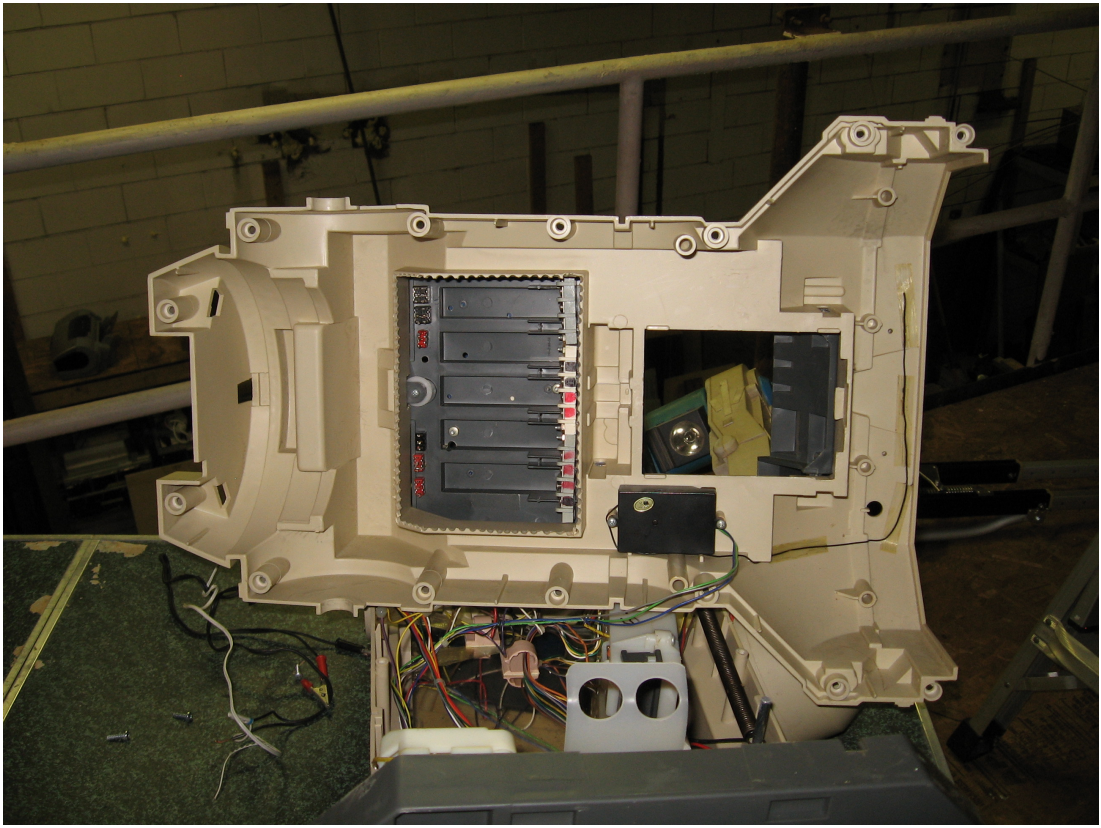

Figure 10: IMG_2125.JPG

Figure 11: IMG_2126.JPG

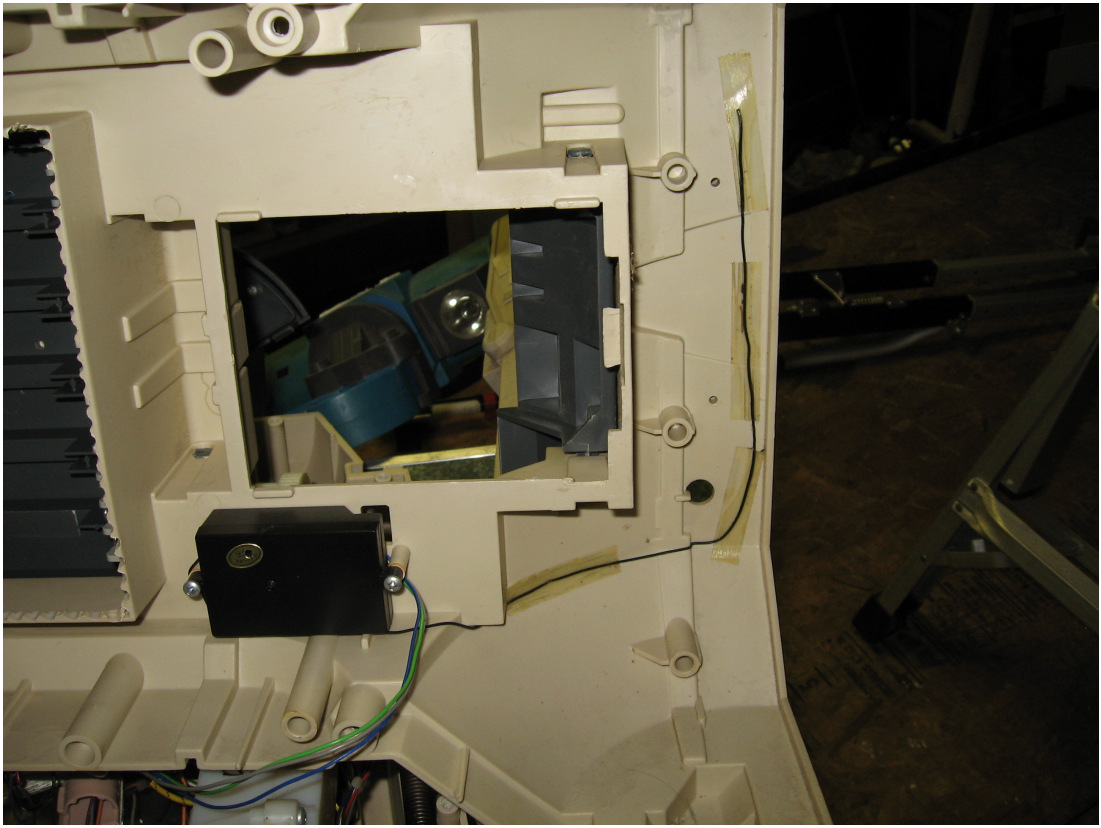

Figure 12: IMG_2127.JPG

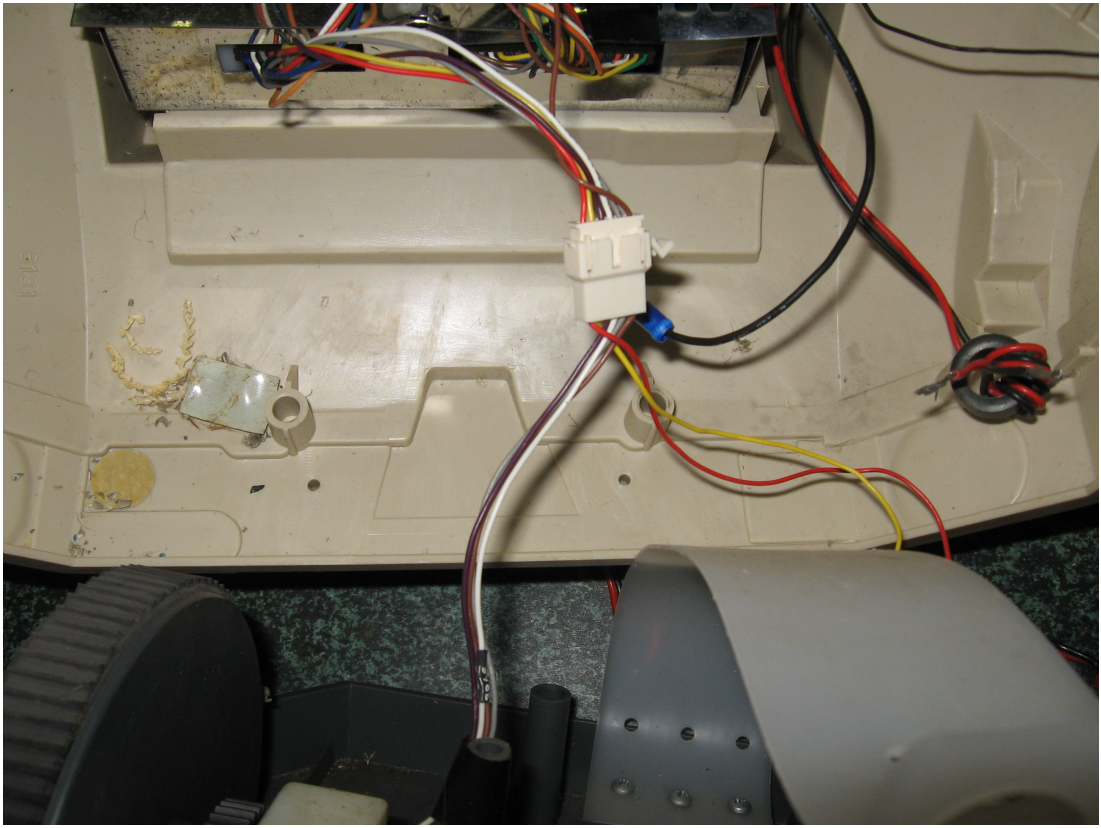

Figure 13: IMG_2128.JPG

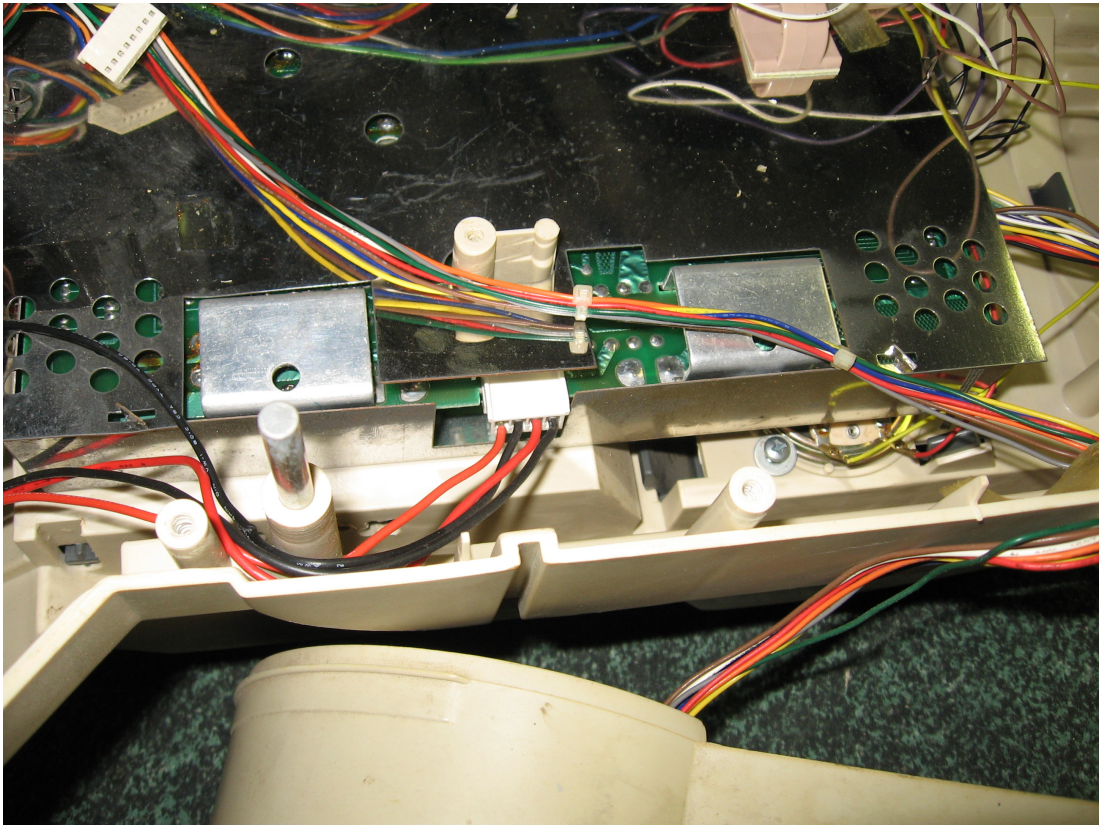

Figure 14: IMG_2129.JPG

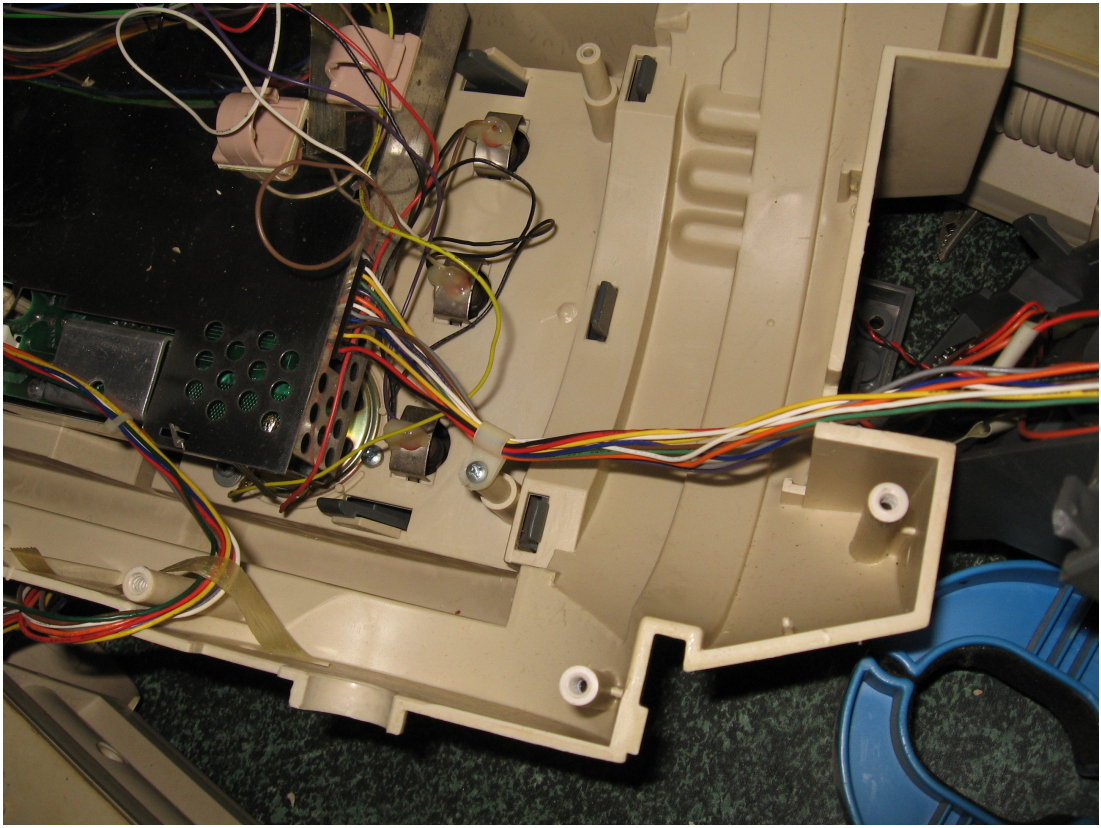

Figure 15: IMG_2130.JPG

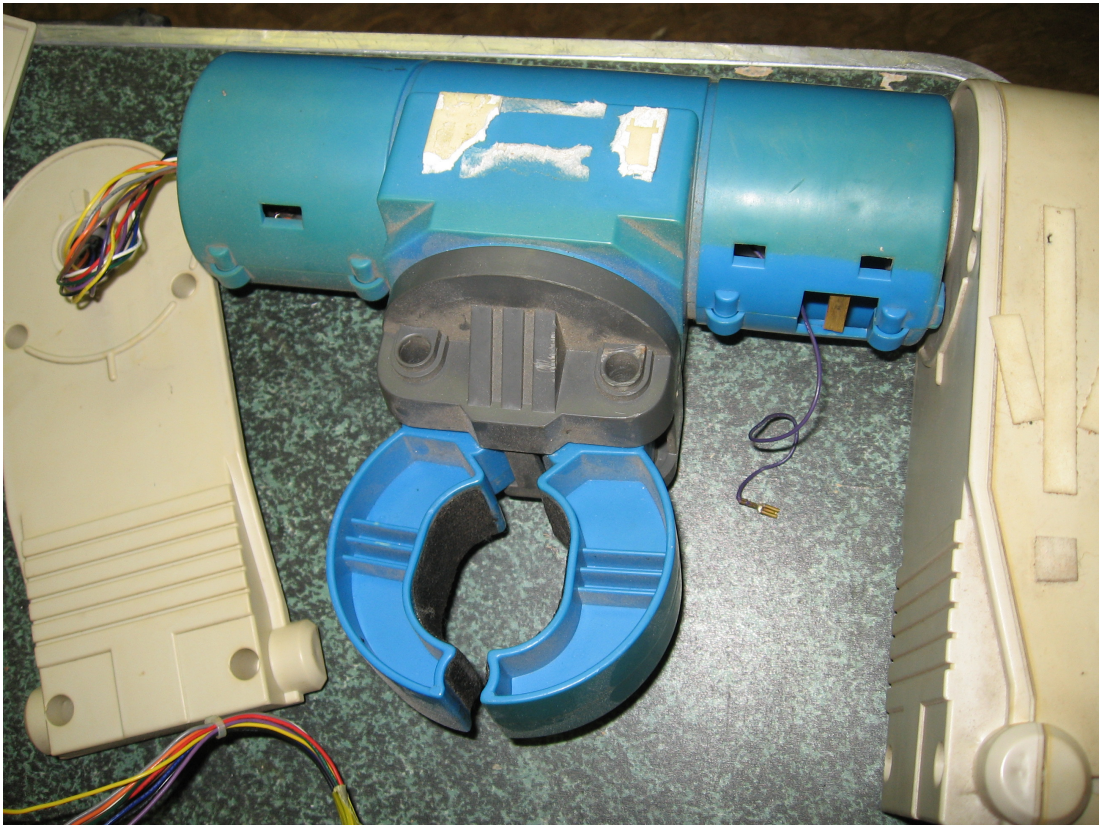

Figure 16: IMG_2131.JPG

Cyberia Makerspace — workshop session

Timestamp photos and hero shot from the teardown session. Canonical `IMG_2116–IMG_2131` frames are in Disassembly above. Archive: `Chassis/Photos/cyberia-makerspace/`.

Figure 17: 2012-09-07_20-53-36_354.jpg

Figure 18: 2012-09-07_20-53-50_564.jpg

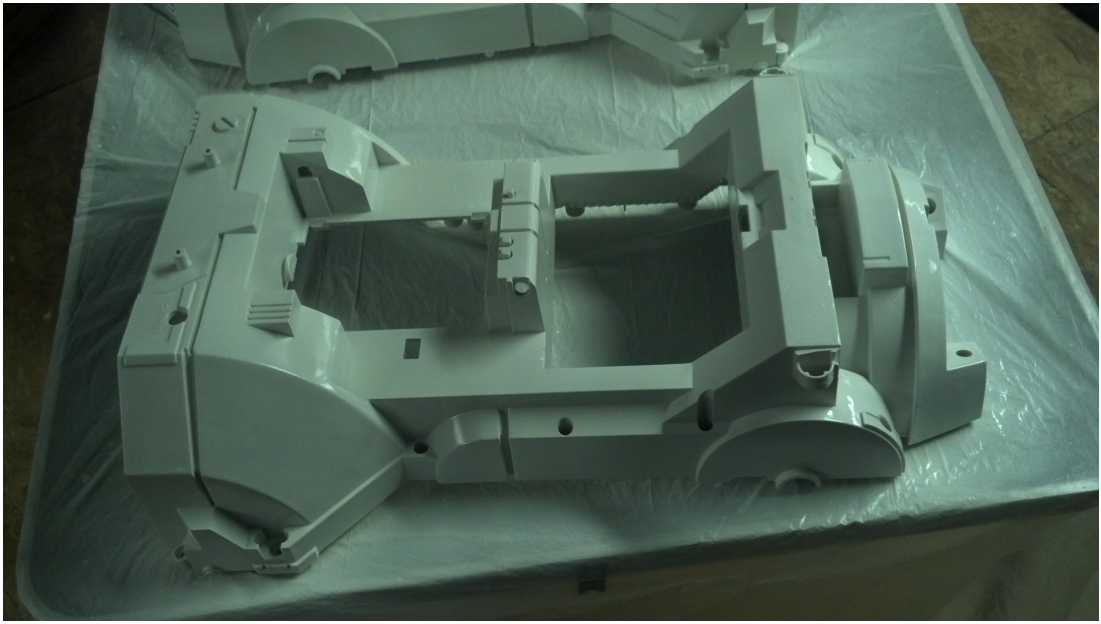

Figure 19: 2012-09-07_20-54-04_577.jpg

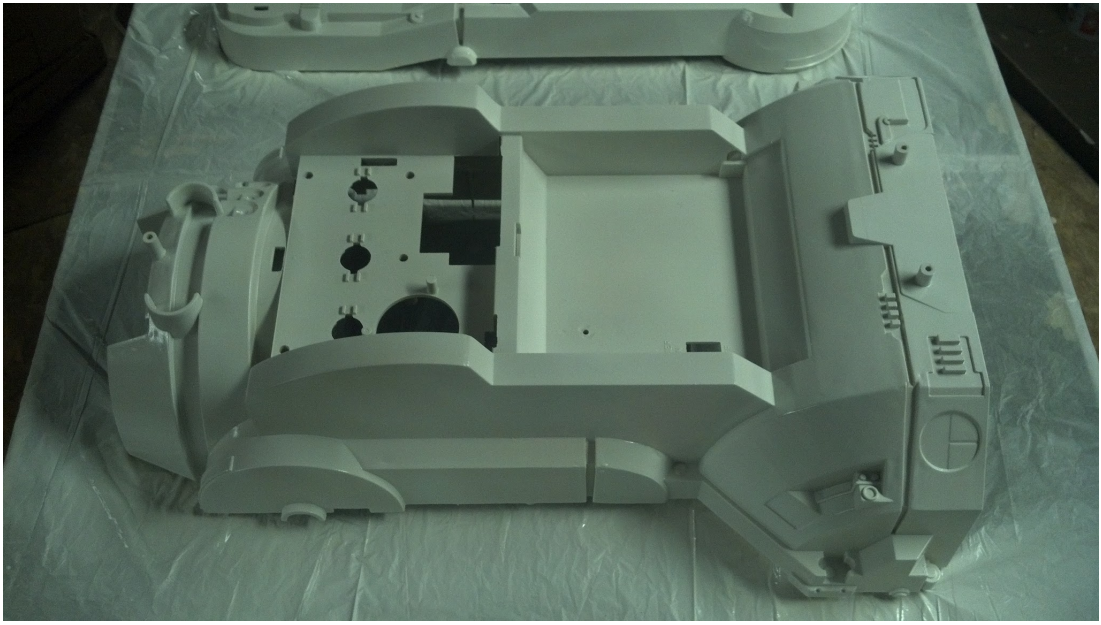

Figure 20: 2012-09-07_20-54-13_422.jpg

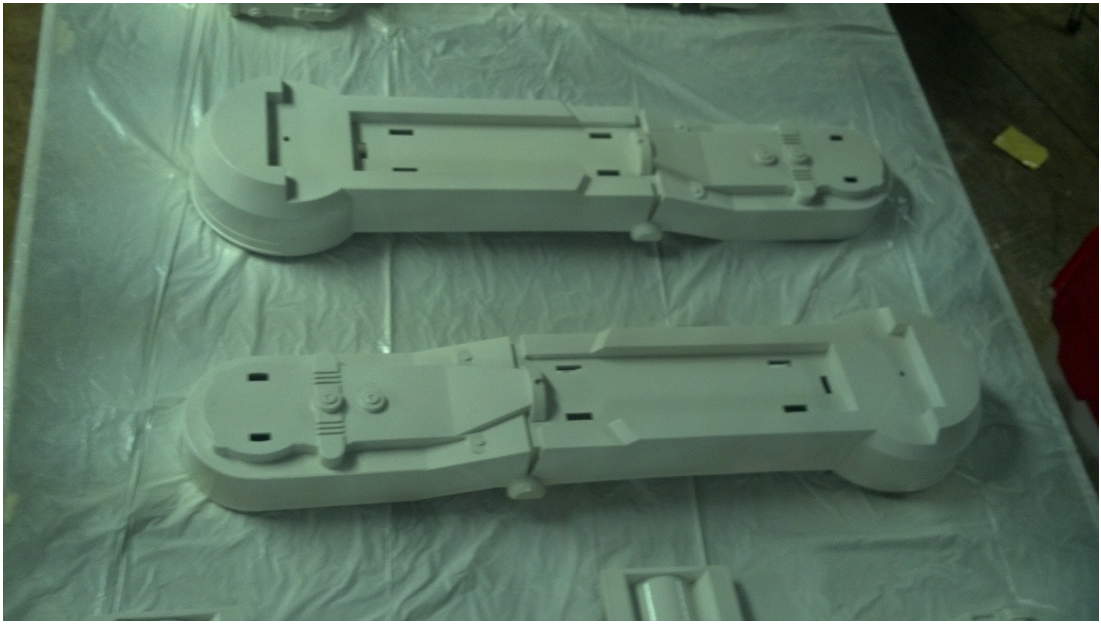

Figure 21: 2012-09-07_20-54-24_707.jpg

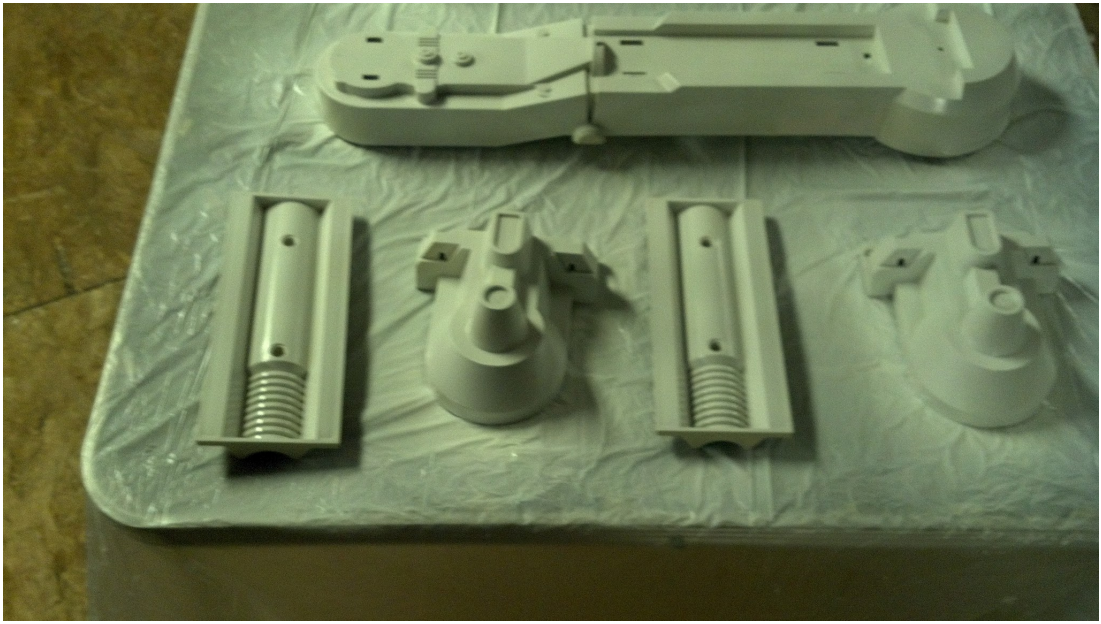

Figure 22: 2012-09-07_20-54-31_57.jpg

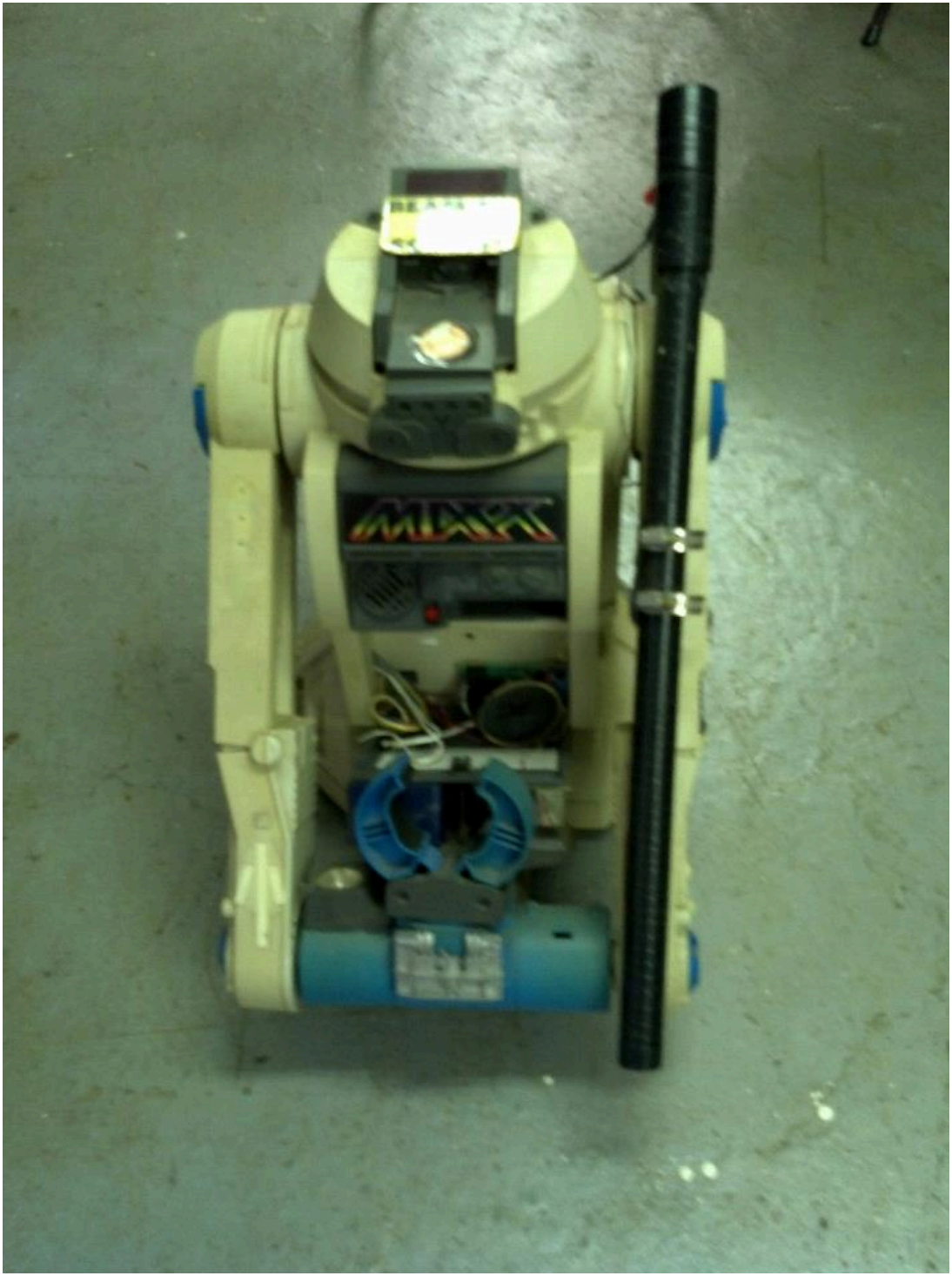

Figure 23: Cyberia-Maxx-Steele.jpg

Internal autopsy

Opened chassis: mainboard, base, arm balance spring, and backoff views. Archive: `Chassis/Photos/autopsy/`.

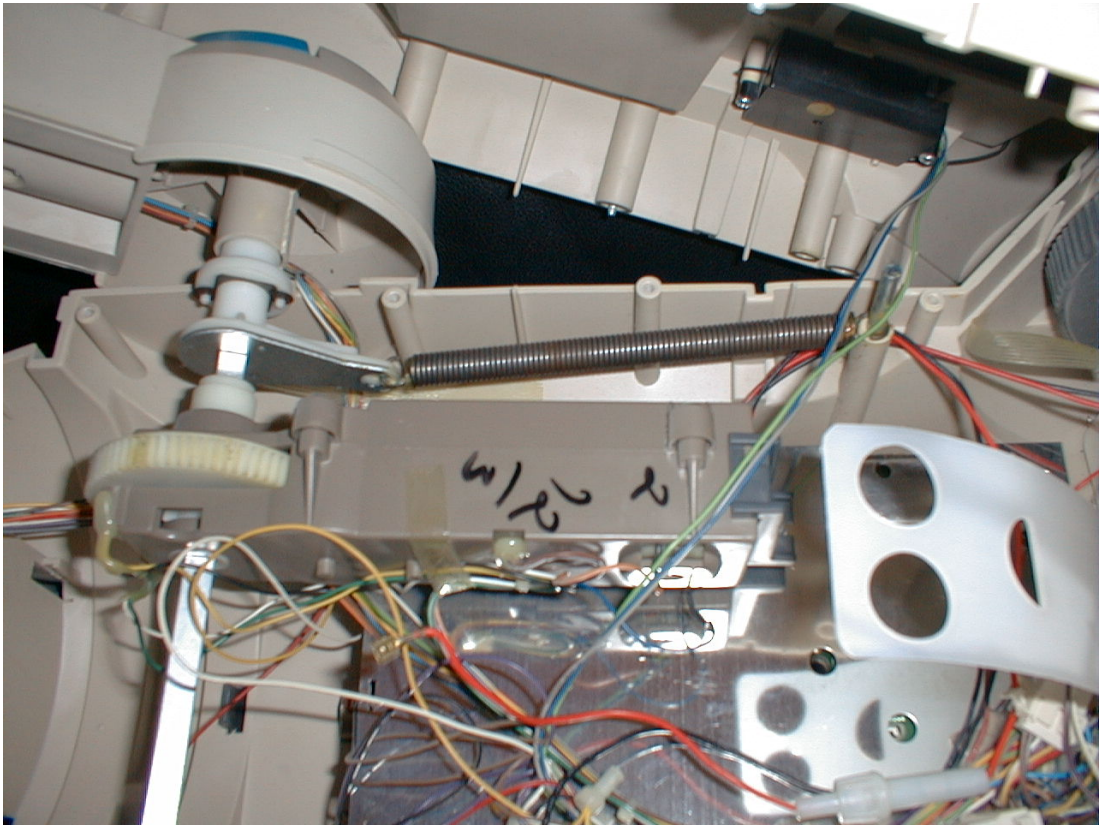

Figure 24: Arm-Balance-Spring.jpg

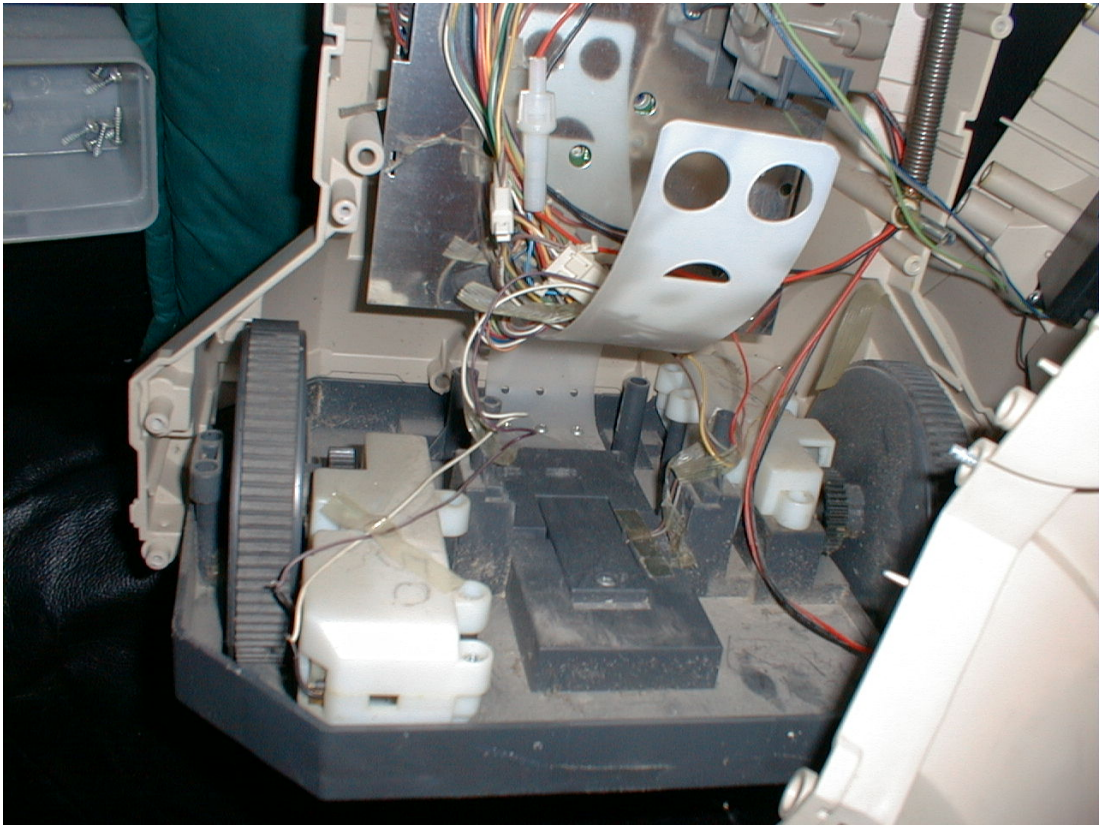

Figure 25: Base-and-battery-mount.jpg

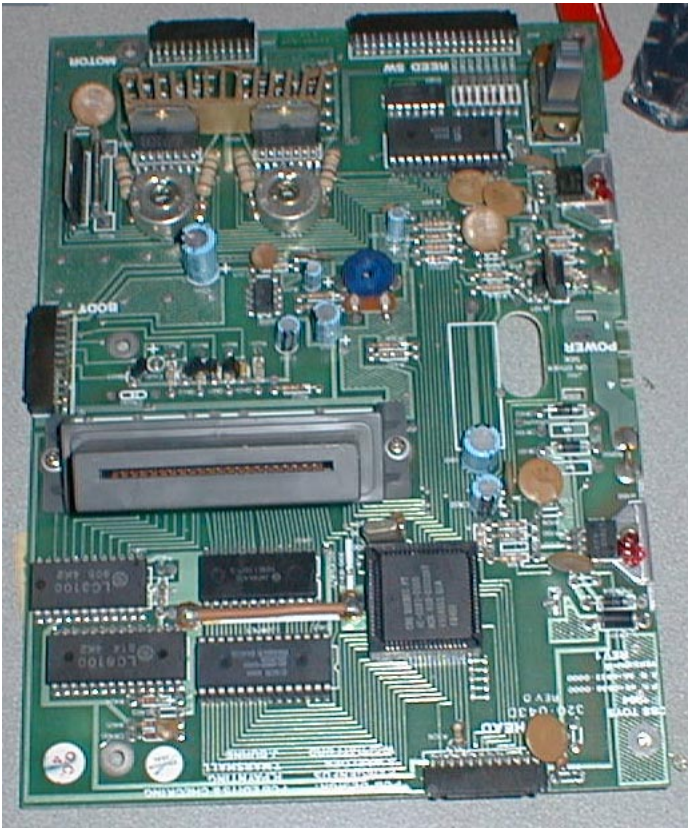

Figure 26: Main-Board-1.jpg

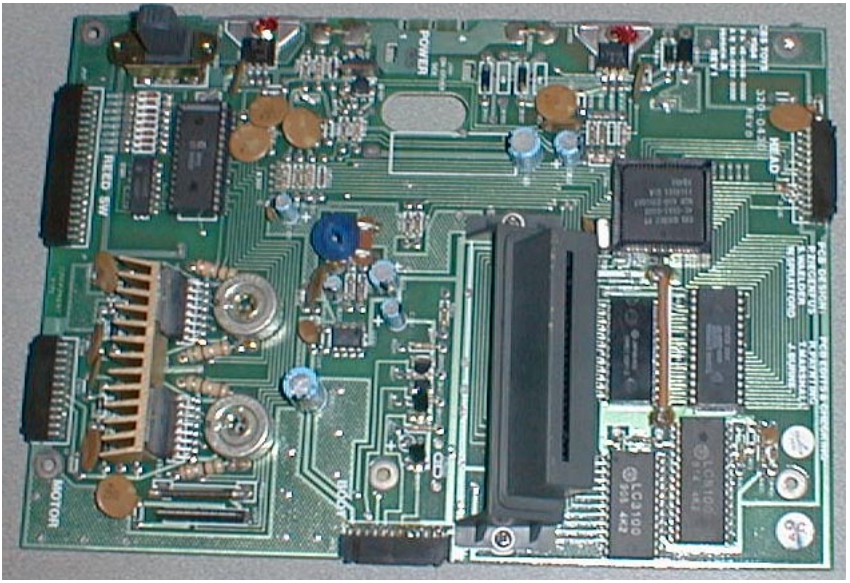

Figure 27: Main-Board.jpg

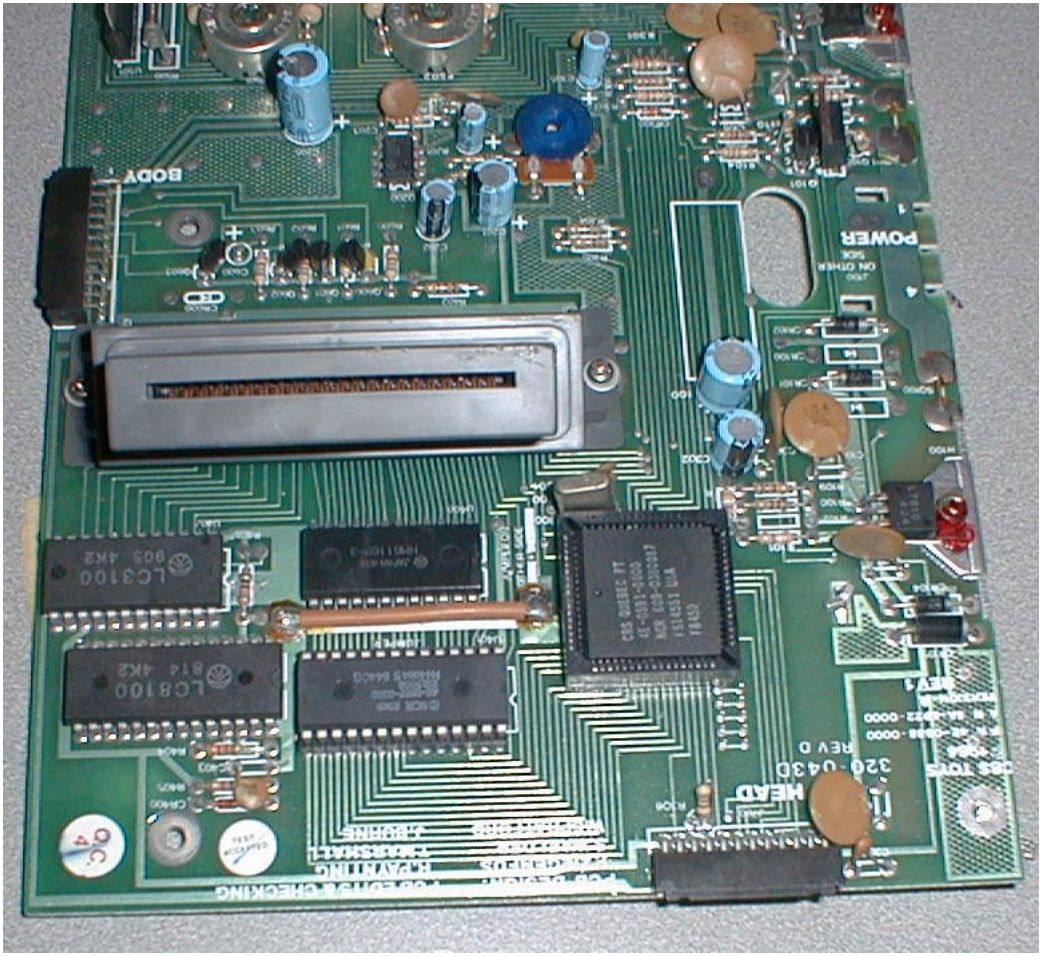

Figure 28: PIC00024.jpg

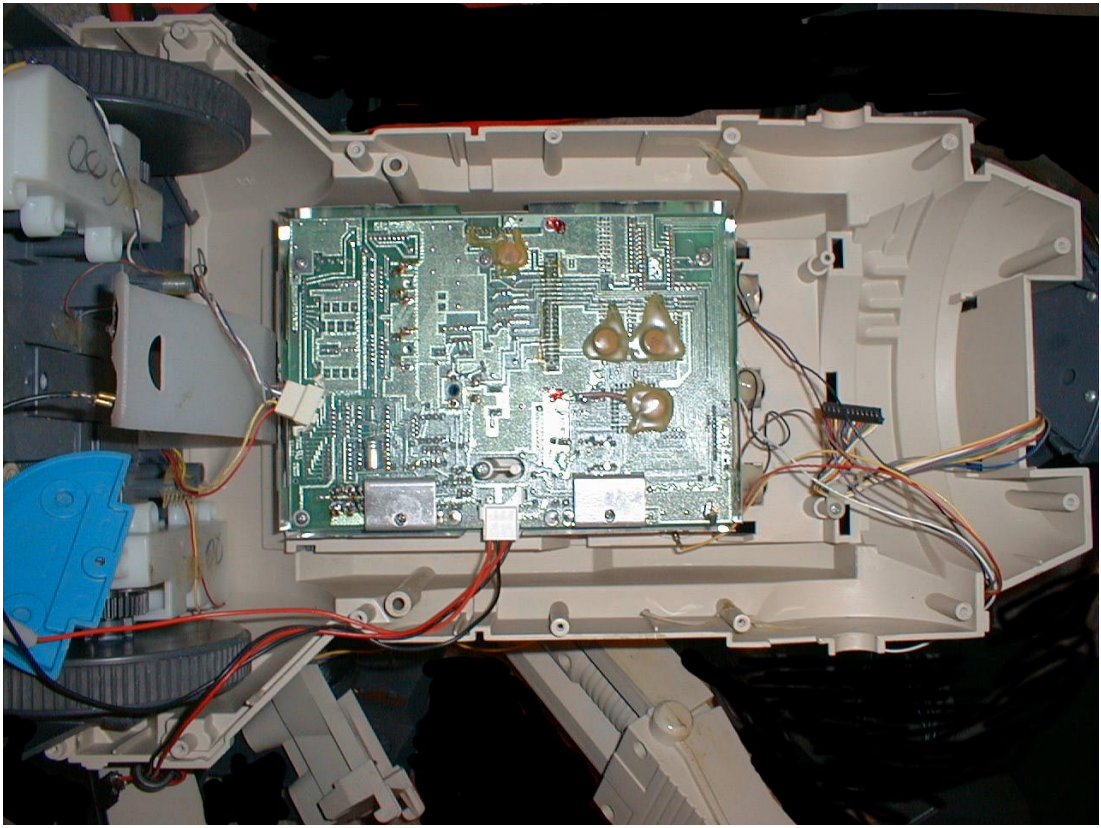

Figure 29: back-off-arms-off.jpg

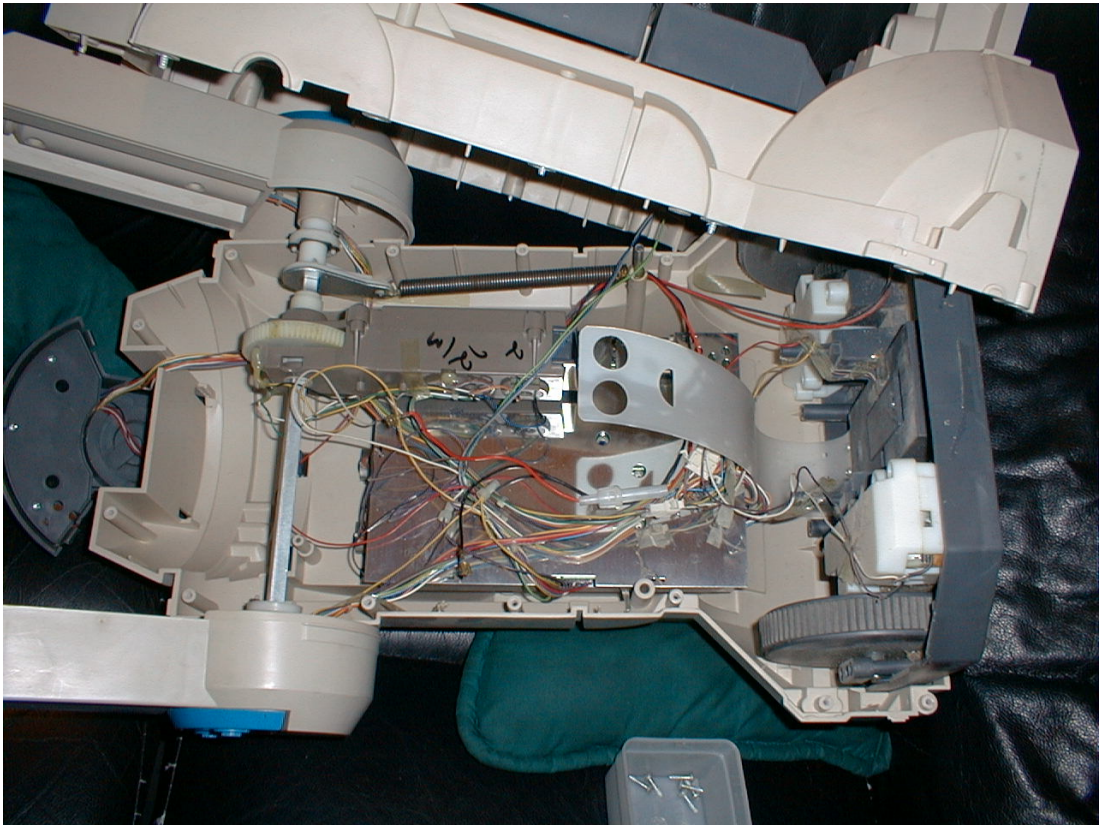

Figure 30: backoff.jpg

Exterior and collection

Mint/vintage toy listing photos: exterior and detail shots. Remote photos: `Transmitter/Photos/Product/`. Archive: `Chassis/Photos/mint-vintage-toys/`.

Figure 31: 05-5.jpg

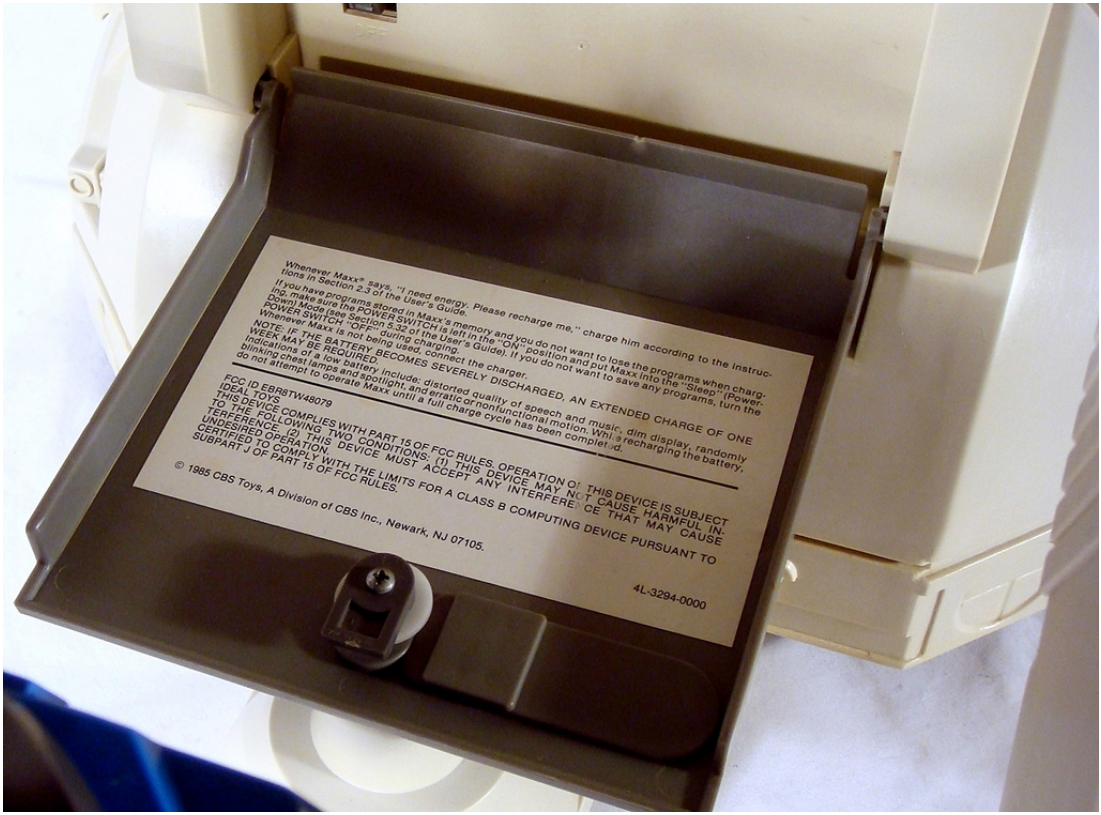

Figure 32: 06-6.jpg

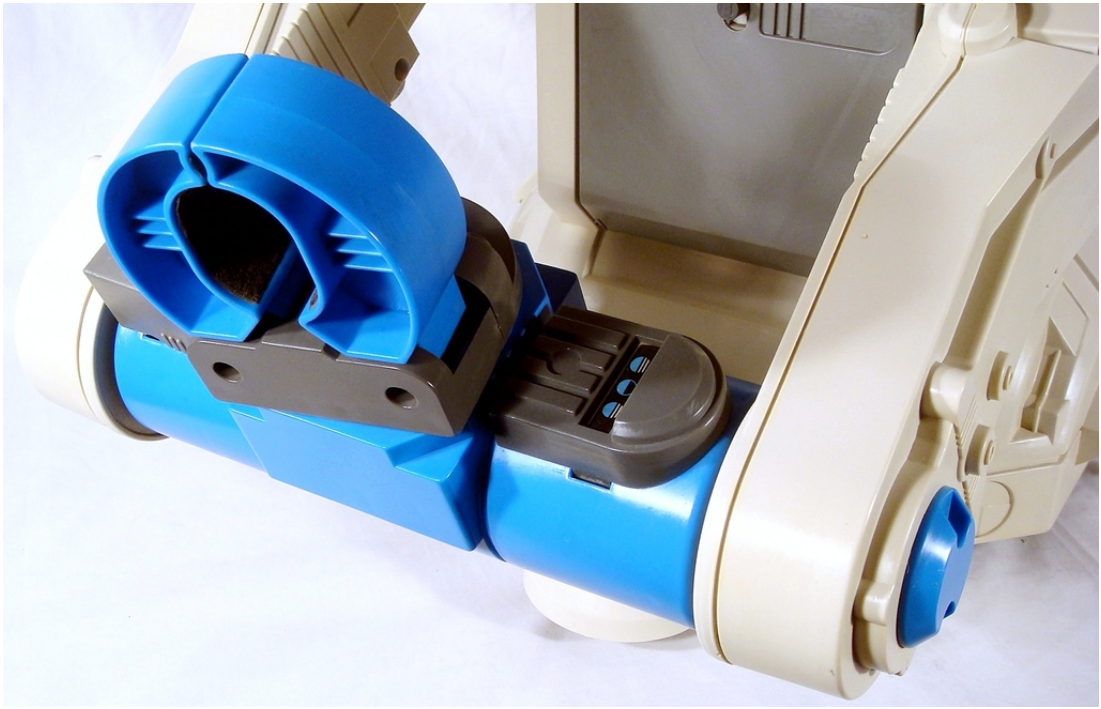

Figure 33: 07-8.jpg

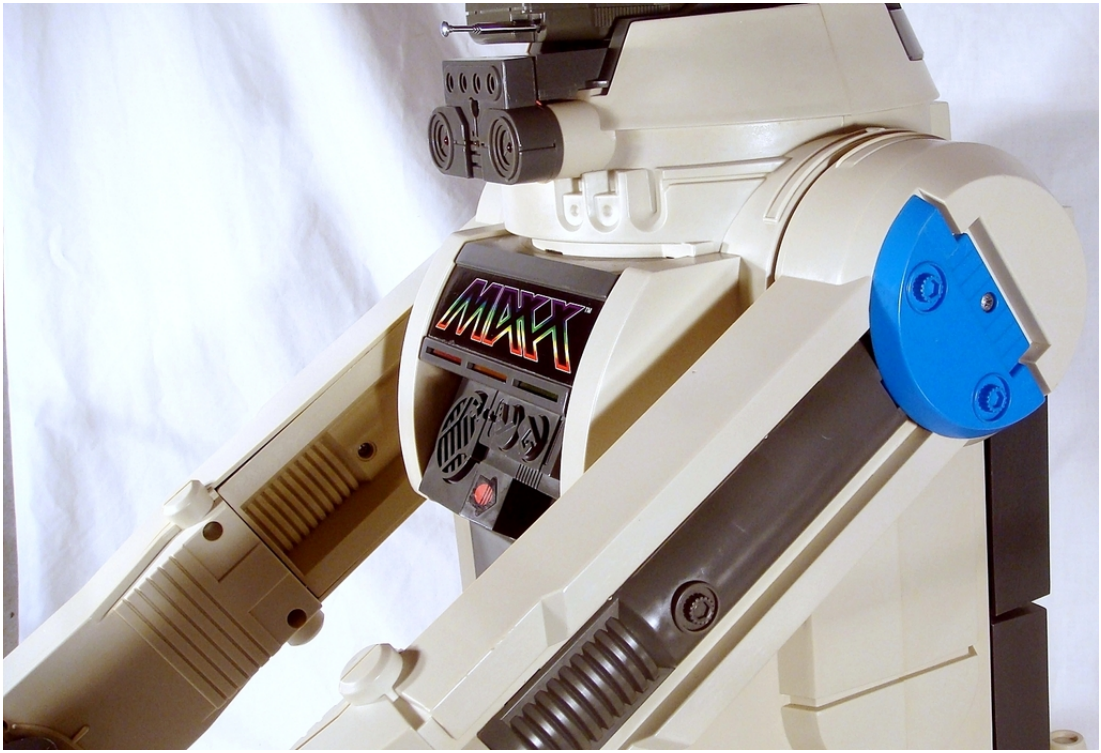

Figure 34: 08-6.jpg

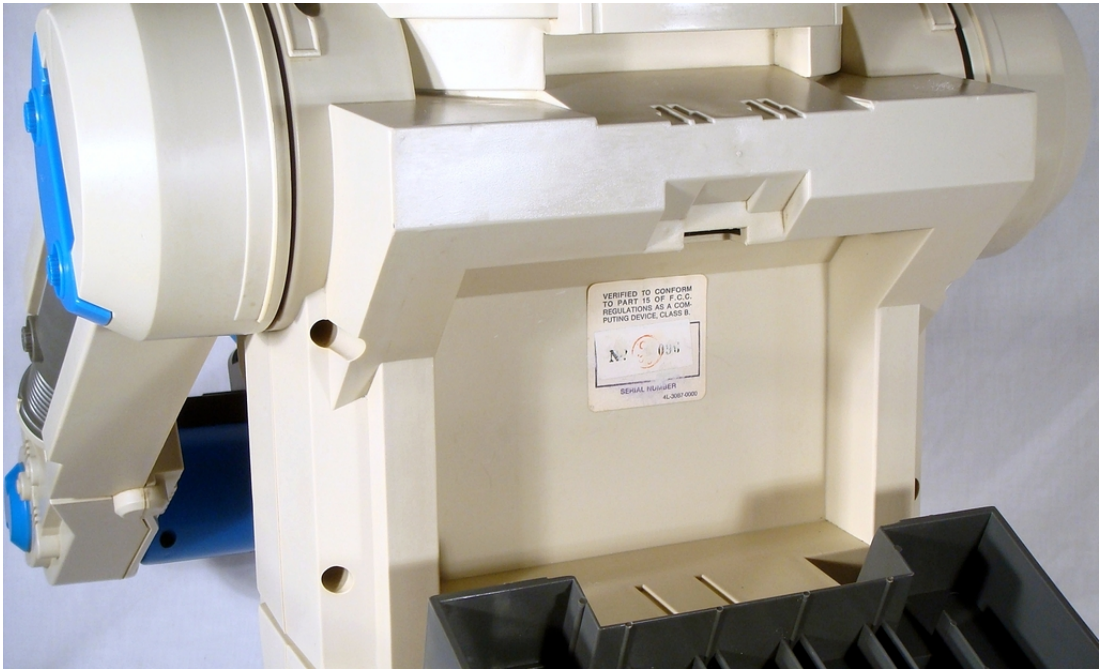

Figure 35: 12-6.jpg

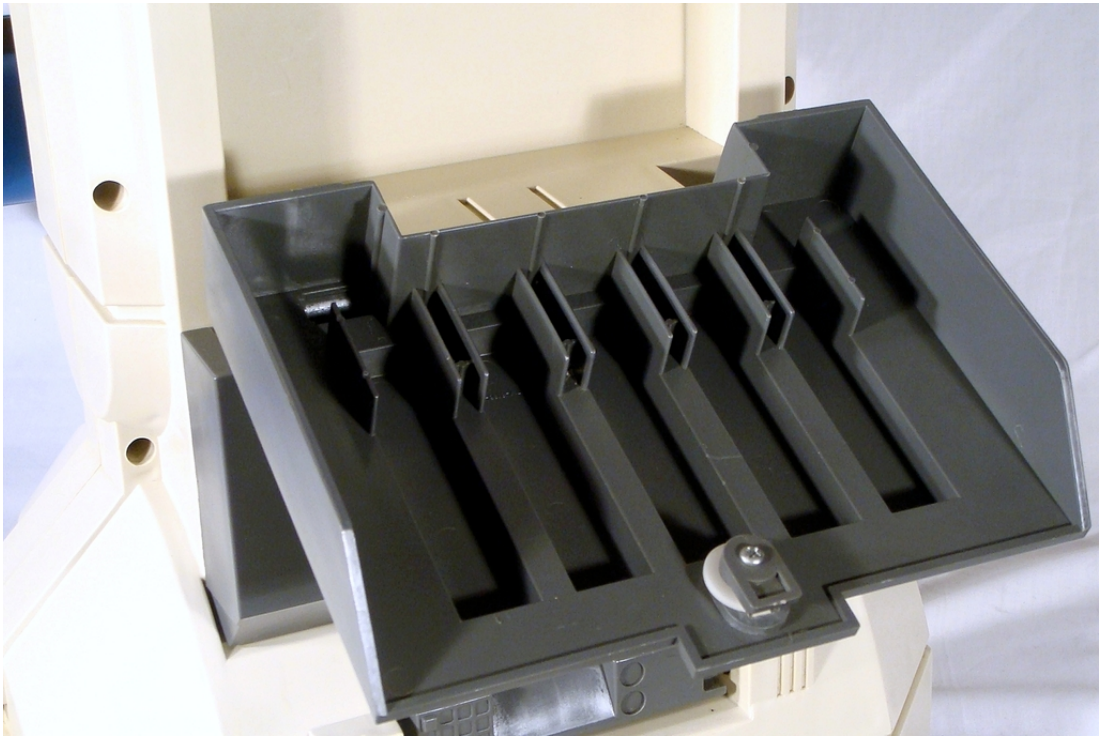

Figure 36: 13-6.jpg

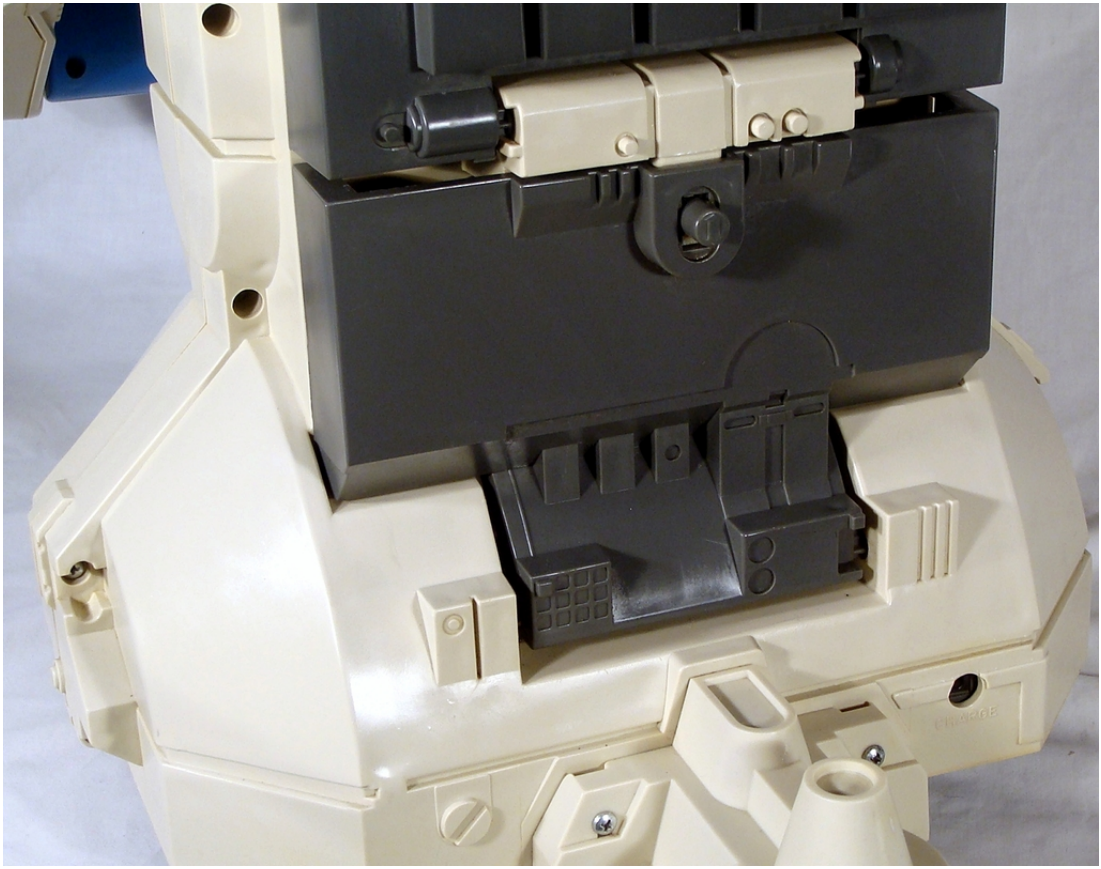

Figure 37: 15-7.jpg

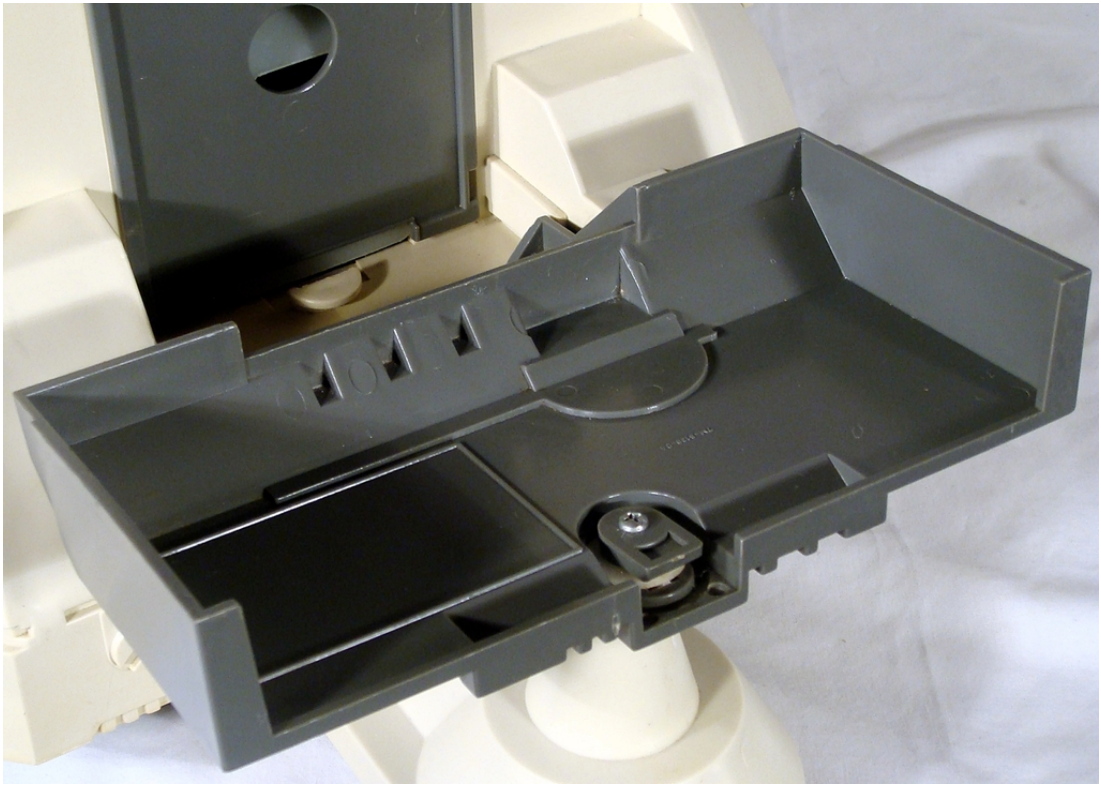

Figure 38: 16-6.jpg

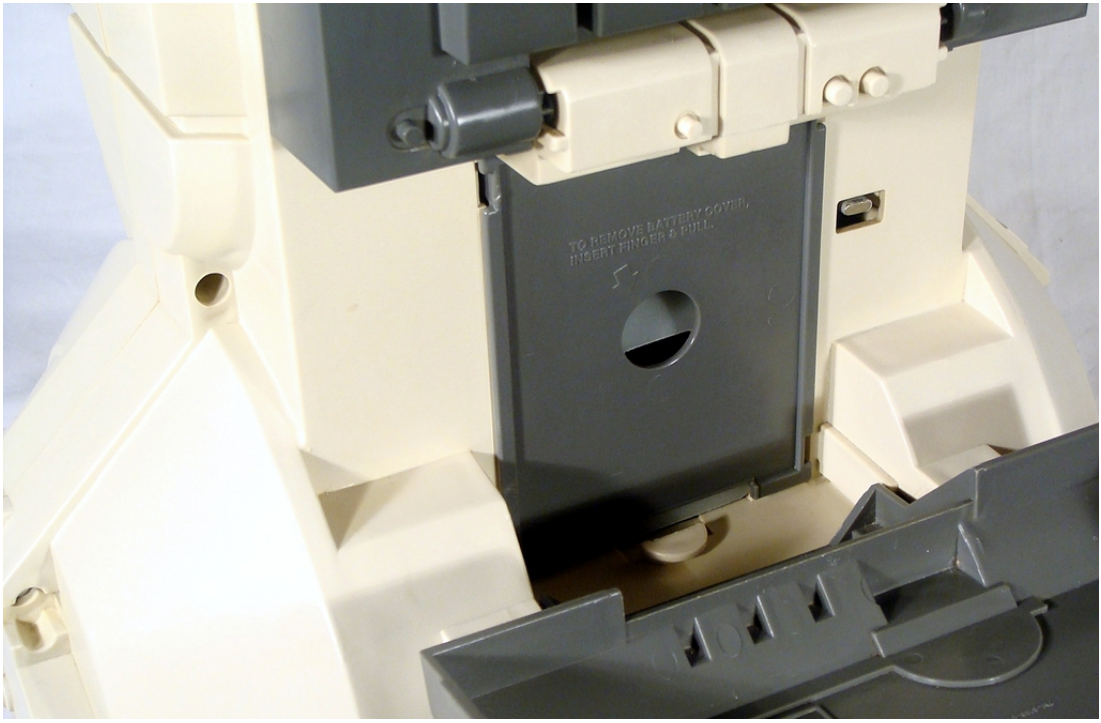

Figure 39: 17-6.jpg

Figure 40: 27-3.jpg

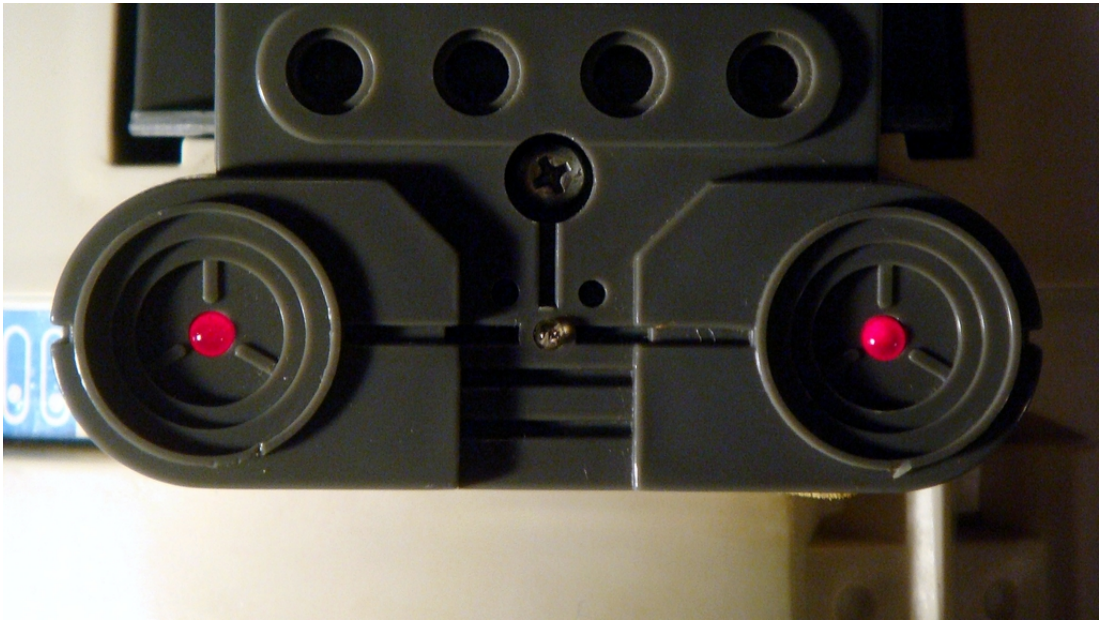

Figure 41: 33-3.jpg

Figure 42: 34-3.jpg

Figure 43: 35-3.jpg

Figure 44: 36-3.jpg

Figure 45: 37-2.jpg

Figure 46: 38-2.jpg

Figure 47: 39-2.jpg

Figure 48: 40-2.jpg

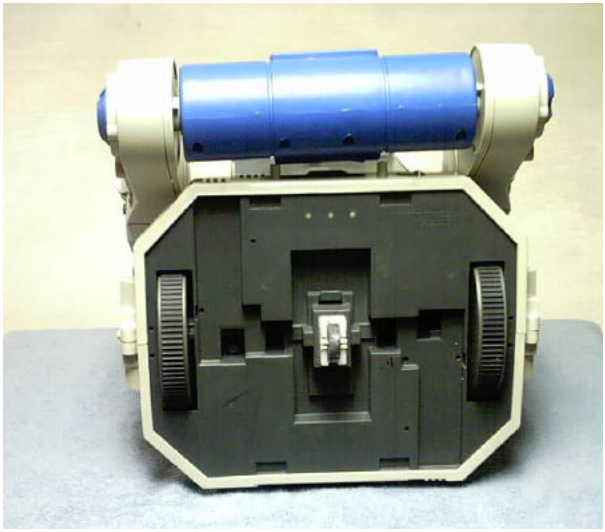

Figure 49: maxx_bottom.jpg

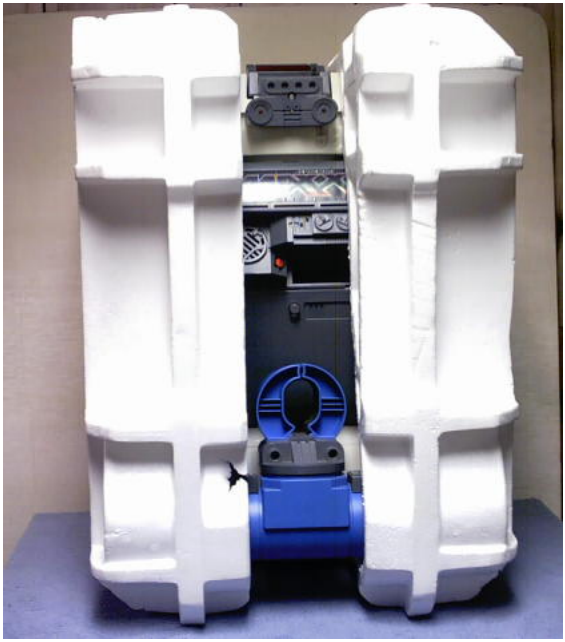

Figure 50: maxx_in_packing.jpg

Figure 51: maxxsteal7.jpg

Case and packaging

Retail case, press, and promotional photos. Archive: `Chassis/Photos/case/`.

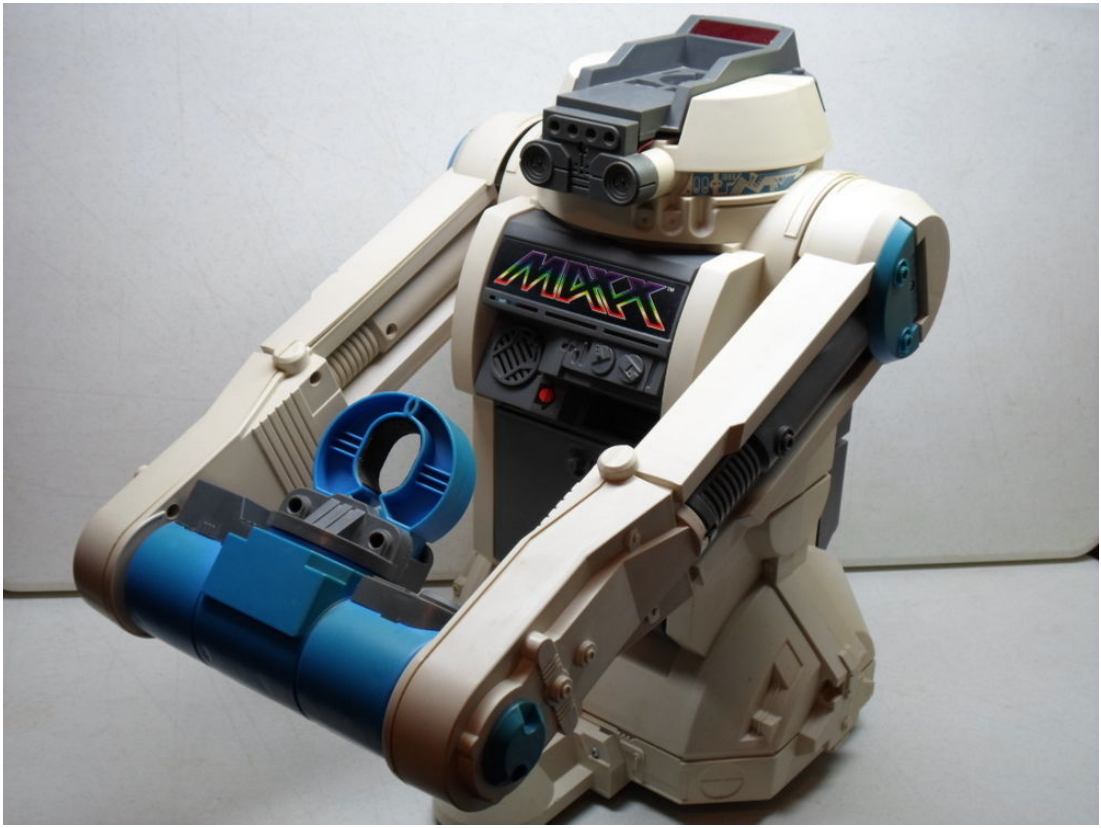

Figure 52: 1000x1000.jpg

Figure 53: Maxx-Steele-Press-Photo.JPG

Figure 54: Robo-Force.JPG

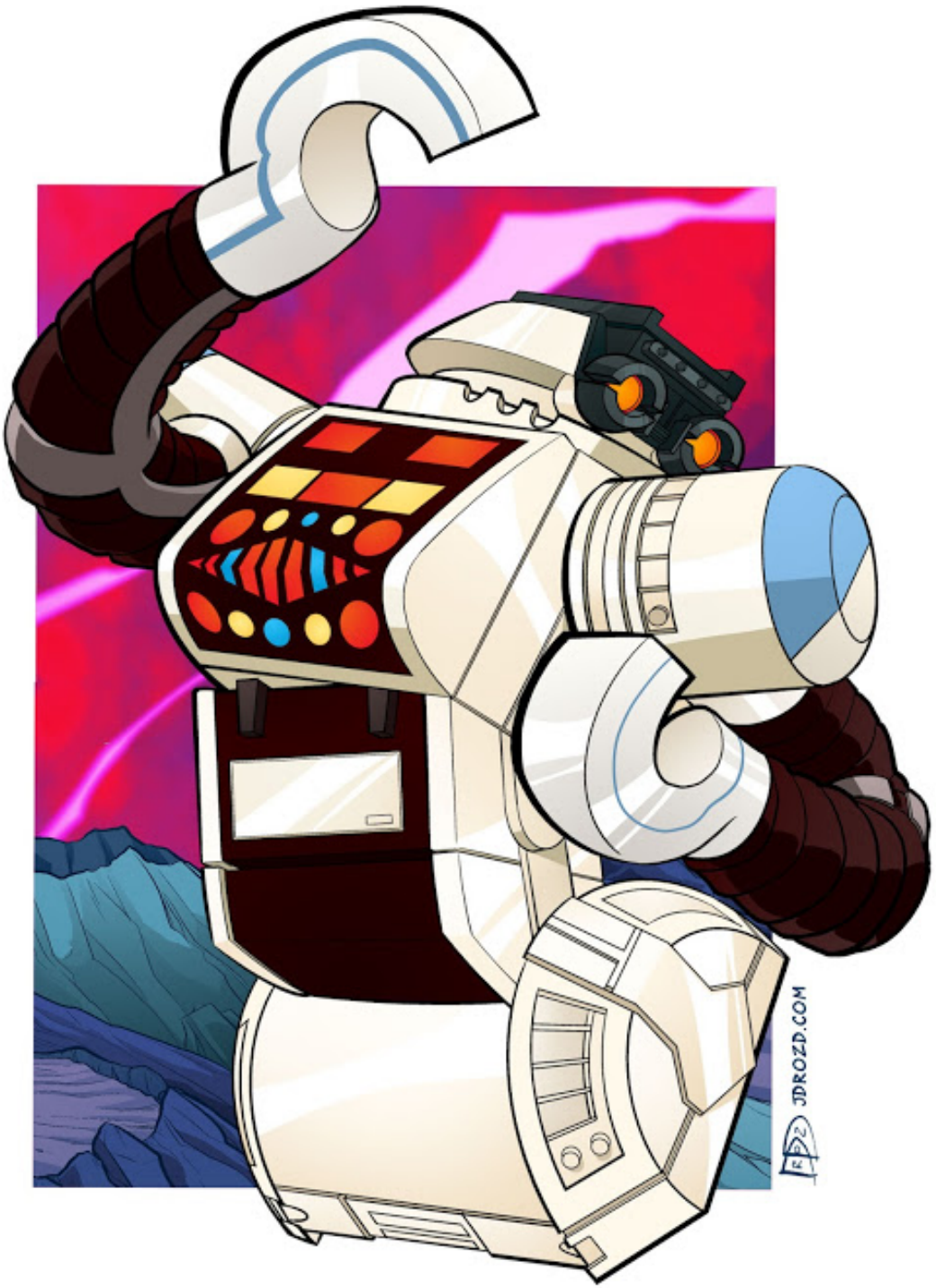

Figure 55: maxx-steele.jpg

Body artwork

Vector logos and body graphics (SVG source files; raster booklet shows filenames). Archive: `Chassis/Artwork/`.

- `Maxx-Steele-MAXX-Logo.svg` — vector artwork (open in repo for full SVG)
- `Maxx-Steele-Neck-Tattoo.svg` — vector artwork (open in repo for full SVG)
- `Maxx-Steele-The-Personal-Robot.svg` — vector artwork (open in repo for full SVG)

Previous: Chapter 2 — Reassembly

Maxx-Steele-1984-Robot

github.com/webaugur/Maxx-Steele-1984-Robot